Experiment – 21

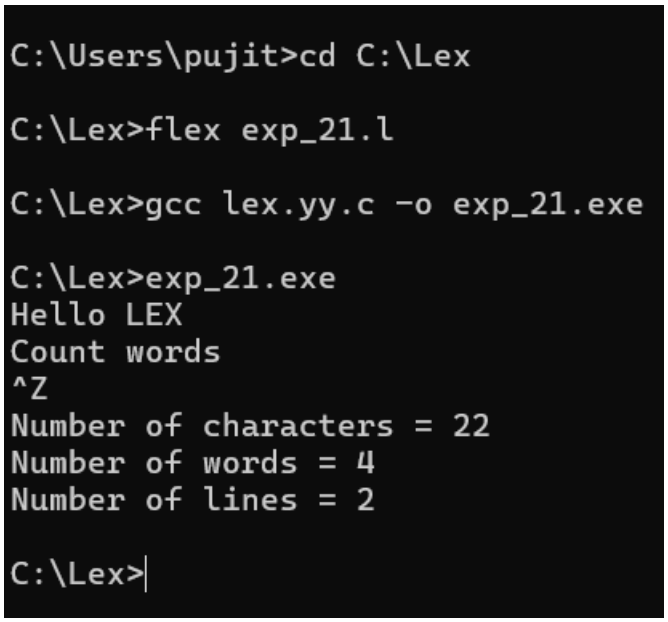
-> LEX CODE :

```
%{
#include <stdio.h>
int chars = 0, words = 0, lines = 0, inword = 0;
%}
%%
[^ \t\n]+  { chars += yyleng; if (!inword) words++; inword = 1; }
[ \t]+     { chars += yyleng; inword = 0; }
\n         { chars++; lines++; inword = 0; }
%%
int main(void) {
    yylex();
    printf("Characters: %d\nWords: %d\nLines: %d\n", chars, words, lines);
    return 0;
}
int yywrap(void) { return 1; }
```

-> INPUT :

```
Hello LEX
Count words
```

-> OUTPUT :



Experiment – 22

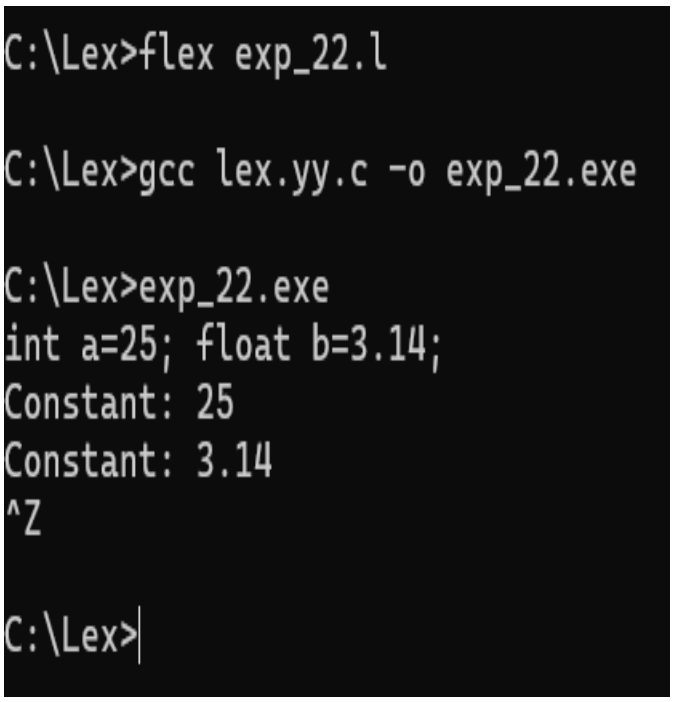
-> LEX CODE :

```
%{
#include <stdio.h>
%}
%%
[0-9]+(\.[0-9]+)?  { printf("Constant: %s\n", yytext); }
.|\n               ;
%%
int main(void) {
   yylex();
   return 0;
}
int yywrap(void) { return 1; }
```

-> INPUT :

int a=25; float b=3.14;

-> OUTPUT :



Experiment – 23

-> LEX CODE :

```
%{
#include <stdio.h>
int defines = 0, includes = 0;
%}
%%
^[ \t]*#[ \t]*define([ \t]+).*   { defines++; }
^[ \t]*#[ \t]*include([ \t]+).* { includes++; }
\n                          ;
.                           ;
%%
```

```
int main(void) {
    yylex();
    printf("#define count: %d\n#include count: %d\n", defines, includes);
    return 0;
}
int yywrap(void) { return 1; }
```

-> INPUT

```
#include <stdio.h>
#define SIZE 10
#define MAX 20
```

-> OUTPUT :

```
C:\Lex>flex exp_23.l

C:\Lex>gcc lex.yy.c -o exp_23.exe

C:\Lex>exp_23.exe
#define count: 0
#include count: 0

C:\Lex>exp_23.exe
#include <stdio.h>
#define SIZE 10
#define MAX 20
^Z
#define count: 2
#include count: 1
```

Experiment – 24

-> LEX CODE :

```
%{
#include <stdio.h>
%}
%%
"<"[^>]*">"   { printf("HTML tag: %s\n", yytext); }
.|\n          ;
%%
int main(void) {
    yylex();
    return 0;
}
int yywrap(void) { return 1; }
```

-> INPUT :

```
<html><body><h1>Hello</h1></body></html>
```

-> OUTPUT :

```
C:\Lex>flex exp_24.l

C:\Lex>gcc lex.yy.c -o exp_24.exe

C:\Lex>exp_24.exe
<html><body><h1>Hello</h1></body></html>
HTML tag: <html>
HTML tag: <body>
HTML tag: <h1>
HTML tag: </h1>
HTML tag: </body>
HTML tag: </html>
^Z
```

Experiment – 25

-> LEX CODE:

```
%{
#include <stdio.h>
int line_no = 1;
%}
%%
.*\n  { printf("%d: %s", line_no++, yytext); }
.+    { printf("%d: %s\n", line_no++, yytext); }
%%
int main(void) {
   yylex();
   return 0;
}
int yywrap(void) { return 1; }
```

-> INPUT :

```
first line
second line
third line
dsadasf
35243
```

-> OUTPUT :

```
C:\Lex>flex exp_25.l

C:\Lex>gcc lex.yy.c -o exp_25.exe

C:\Lex>exp_25.exe
first line
1: first line
second line
2: second line
third line
3: third line
first line
4: first line
dsadasf
5: dsadasf
35243
6: 35243
```

Experiment – 26

-> LEX CODE :

```
%{
#include <stdio.h>
%}
%x COMMENT
%%
\"([^\\\"]|\\.)*\"    { ECHO; }
\'([^\\\']|\\.)*\'    { ECHO; }
"/*"                  { BEGIN(COMMENT); }
<COMMENT>"*/"         { BEGIN(INITIAL); }
<COMMENT>\n           { ECHO; }
```

```
<COMMENT>.          ;
"//".*              ;
.|\n                { ECHO; }
%%
int main(int argc, char **argv) {
    if (argc != 3) {
        fprintf(stderr, "Usage: %s input.c output.c\n", argv[0]);
        return 1;
    }
    yyin = fopen(argv[1], "r");
    if (!yyin) {
        perror(argv[1]);
        return 1;
    }
    yyout = fopen(argv[2], "w");
    if (!yyout) {
        perror(argv[2]);
        fclose(yyin);
        return 1;
    }
    yylex();
    fclose(yyin);
    fclose(yyout);
    return 0;
}
int yywrap(void) { return 1; }
```

-> INPUT :

input.c

```
int main() {
    int a = 1; // This is a comment

    /* This is a
       multi-line comment */

    printf("/* keep string */");
    return 0;
}
```

-> OUTPUT :

```
C:\Lex>flex exp_26.l

C:\Lex>gcc lex.yy.c -o exp_26.exe

C:\Lex>exp_26.exe input.c output.c

C:\Lex>
```

```
int main() {
    int a = 1;



    printf("/* keep string */");
    return 0;
}
```

Experiment – 27

-> LEX CODE :

```
%{
#include <stdio.h>
%}
%%
[A-Z]+  { printf("Uppercase word: %s\n", yytext); }
.|\n    ;
%%
int main(void) {
    yylex();
    return 0;
}
int yywrap(void) { return 1; }
```

-> INPUT :

```
HELLO world LEX TEST123 ABC
```

-> OUTPUT :

```
C:\Lex>flex exp_27.l

C:\Lex>gcc lex.yy.c -o exp_27.exe

C:\Lex>exp_27.exe
HELLO world LEX TEST123 ABC
Uppercase word: HELLO
Uppercase word: LEX
Uppercase word: TEST
Uppercase word: ABC
^Z
```

Experiment – 28

-> LEX CODE :

```
%{
#include <stdio.h>
%}
%%
[A-Za-z0-9._%+-]+@[A-Za-z0-9.-]+\.[A-Za-z]{2,}   { printf("Accepted\n"); }
.+                                               { printf("Not Accepted\n"); }
\n                                               ;
%%
int main(void) {
    printf("Enter email: ");
    yylex();
    return 0;
}
int yywrap(void) { return 1; }
```

-> INPUT :

pujitha123@gmail.com pujitha@gmail

-> OUTPUT :

```
C:\Lex>flex exp_28.l

C:\Lex>gcc lex.yy.c -o exp_28.exe

C:\Lex>exp_28.exe
Enter email: pujitha123@gmail.com
Accepted
pujitha@gmail
Not Accepted
^Z
```

Experiment – 29

-> LEX CODE :

```
%{
#include <stdio.h>
%}
%%
abc       { printf("ABC"); }
.|\n      { ECHO; }
%%
int main(void) {
    yylex();
    return 0;
}
int yywrap(void) { return 1; }
```

-> INPUT :

```
abcdefghabcijkla
```

-> OUTPUT :

```
C:\Lex>flex exp_29.l

C:\Lex>gcc lex.yy.c -o exp_29.exe

C:\Lex>exp_29.exe
abcdefghabcijkla
ABCdefghABCijkla
^Z
```

Experiment – 30

-> LEX CODE :

```
%{
#include <stdio.h>
%}
%%
[1-9][0-9]{9}   { printf("Mobile Number Valid\n"); }
[^\n]+         { printf("Mobile Number Invalid\n"); }
\n             ;
%%
int main(void) {
    printf("Enter Mobile Number: ");
    yylex();
    return 0;
}
int yywrap(void) { return 1; }
```

-> INPUT :

7856453489

-> OUTPUT :

```
C:\Lex>flex exp_30.l

C:\Lex>gcc lex.yy.c -o exp_30.exe

C:\Lex>exp_30.exe
Enter Mobile Number: 1234565678
Mobile Number Valid
32142534675867 98
Mobile Number Invalid
```